SAY WATT ROBOTICS · FIELD LOG

Week 2.1 — 100 Cycles, Zero Leaks — Phase 2 Opens with the Float Rubric and a Syringe Test

Sunday, September 20, 2026 · 3:00 – 5:00 PM · 5 students

Five of us at 2.1 — Trenton, Isaac, Tyler, Rhylan, and Rebecca — from 3:00 to 5:00, for the first session of Phase 2. Small table, big agenda. The first piece of the 2027 game is out: MATE has released the preview mission for the float, and the team went through its rubric line by line. The 2027 float has to stay under 60 centimeters tall including its antenna and under 18 across. It has to sink to two meters and hold there for thirty seconds, keep going to four meters, then sample water temperature and light every half meter on the way back up and radio the data home. A buoyancy engine — something that moves water in and out to change the float's density — is worth 20 of the 70 points, and touching the bottom costs five. That is a real engineering problem, and now the team knows exactly what it is. Meanwhile the payload track ran the test the manipulator trade study was waiting on: is a syringe hydraulic system tight enough to drive the claw? They filled the syringes with green-dyed water so any leak would show, then cycled them — 20 push-pull cycles in air, 30 with a syringe fully underwater, and 50 with a plunger submerged in glycerin. A hundred cycles, no leaks. The catch they found was air. Bubbles in the line make the system spongy, and the longer the tube between syringes, the harder it is to bleed, which means a hydraulic manipulator will be hard to set up the same way twice. That is an honest result: it backs the leakage score in the trade table and names setup as the real risk. On the revised trade table the two leading designs came out dead even — linear actuator 29, all-hydraulic 29, the combination 19. The next test is the one that counts — under the water pressure the manipulator will actually see.

5	100	70	20
STUDENTS — THE FIRST SESSION OF PHASE 2	PUSH-PULL CYCLES WITHOUT A LEAK	POINTS IN THE 2027 FLOAT MISSION	OF THEM FOR USING A BUOYANCY ENGINE

WHAT WE LEARNED

- Read the rubric before you design anything. Every one of the float's 70 points traces back to a line in the preview mission — the hold, the profile, the nine readings, the packets, the graphs. A feature that can't point to the line it scores is decoration.
- A buoyancy engine changes the float's density by moving water in and out — not by pushing against the water with a propeller. That distinction is worth 20 points.
- Why a sealed tube needs a way to open: a battery that heats up or vents gas can raise the pressure inside a closed housing, and if the inside pushes harder than the outside, the housing has to open on purpose instead of failing on its own. For 2027 MATE accepts an O-ring end cap that can pop off, at least 2.5 cm across, or a commercial relief valve set at 15 psi or less.
- Color the water. Green dye turns a leak test from "I think it's dry" into something you can see.
- A test can pass and still find the problem. No leaks in 100 cycles — but the same test showed that trapped air, and long tubes that are hard to purge, are the real enemy of a hydraulic manipulator.
- Test in the conditions you will compete in. Surface results are a start; the manipulator works at depth, so the next leak test has to happen under water pressure.

DECISIONS MADE

- The float mission is the season's first known target. The 2027 preview rubric now drives the design: under 60 cm tall, under 18 cm across, a 2.0 m hold for 30 seconds, a profile to 4.0 m, temperature and light every half meter on the way up, and the data radioed home.
- A buoyancy engine is the plan, not an option. It is worth 20 of the 70 points, and a float that changes depth any other way gives half of them away.
- Syringe hydraulics passed the leak test — 100 cycles across air, underwater, and glycerin with no leakage — which backs the leakage score in the manipulator trade table. Trapped air, and how hard it is to bleed out, is now the risk to beat. On the revised table, linear actuator and all-hydraulic sit tied at 29.
- The next hydraulic test happens under real water pressure. A syringe that holds at the surface hasn't been proven at depth yet.

ACTION ITEMS

- Build the profiler bill of materials
  Management · by 2.2
- Go over the profiler rubric again at the next meeting
  Management · by 2.2
- Research a pressure relief valve for the profiler
  Mechanical group · by 2.2
- Bring the manipulator actuation trade to the full team for review
  Mechanical group · by 2.2
- Research light, pressure, and temperature sensors for the profiler
  Sensor group · by 2.2
- Decide how the profiler talks to the topside
  Software group · by 2.2

STUDENT LEARNING OUTCOMES

- Reviewed the 2027 MATE float preview rubric and translated it into design requirements.
- Designed and ran a 100-cycle leak test on a syringe hydraulic system in air, underwater, and in glycerin, using dyed water to make leaks visible.
- Identified trapped air and tube length as the main risks of a hydraulic manipulator, and fed the result back into the actuation trade table.
- Learned why the float's sealed housing needs a pressure-relief mechanism and what MATE accepts for 2027.
- Set up the profiler's research tasks: bill of materials, pressure relief, sensors, and communications.

LOOKING AHEAD

The profiler gets a bill of materials, and the rubric gets a second pass with more of the team in the room. Research runs on three fronts — a pressure relief valve, the light, pressure, and temperature sensors, and how the float talks to the topside. The manipulator trade goes to the full team for review, and the next hydraulic test moves under real water pressure.

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